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SCC.160: Fundamentals of Communication Systems Tutorial 5

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New timing for the Quiz

TIME: 2023/11/15 14:00-14:30

LOCATION: SX503 and SX606

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Explain the difference between NRZ-L and NRZ-I.

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- **NRZ-L** NRZ-L：用负电压表示1，用正电压表示0
- Non-return-to-zero-level (NRZ-L) is a data encoding scheme in which a negative voltage is used to represent binary one and a positive voltage is used to represent binary zero. It maintains a constant voltage pulse for the duration of a bit time.
- Let us consider

0 1 0 0 1 1 1 0

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NRZ-I

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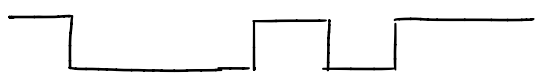
NRZ-I：1表示改变电平，0表示不改变电平

- As with NRZ-L, NRZ-I maintains a constant voltage pulse for the duration of a bit time. The data themselves are encoded as the presence or absence of a signal transition at the beginning of the bit time. **A transition (low to high or high to low) at the beginning of a bit time** denotes a binary 1 for that bit time; no transition indicates a binary 0.
- Let us consider

0 1 0 0 1 1 1 0



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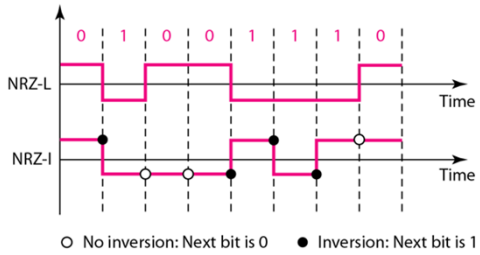

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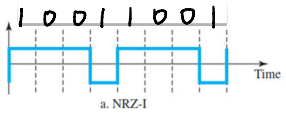
Find the 8-bit data stream for each case depicted in the following figure. For differential Manchester, there is no transition at the beginning of the first bit.



In NRZ-I (NRZ-Invert), a transition (low to high or high to low) at the beginning of a bit time denotes a binary 1 for that bit time; no transition indicates a binary 0.

1表示改变电平,
0表示不改变,维持上一个电平

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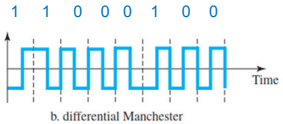
Cont'd

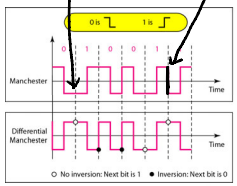
第一个位开头没有过渡

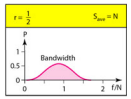
- For differential Manchester, there is no transition at the beginning of the first bit.

比特中间的过渡仅用于同步

In differential Manchester, the transition at the middle of the bit period is used only to provide synchronization. The bit values are represented by the presence of a transition at the beginning of the bit.







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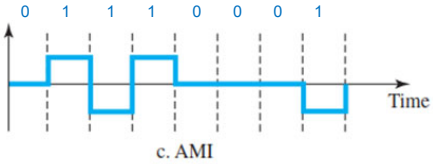
Cont'd AMI: 0 表示零电压
1 表示交替的正负电压

- In AMI, a zero voltage represents binary 0. Binary 1s are represented by alternating positive and negative voltages.



Pseudoternary: 0 表示交替的正负电压
(拟三元) 1 表示零电压

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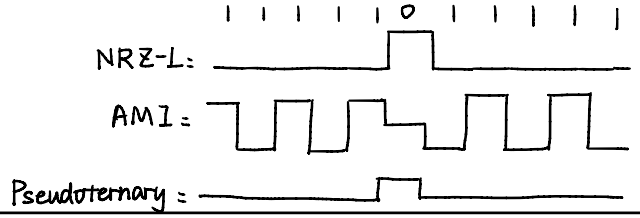

Question 3

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Consider a stream of binary data consisting of a long sequence of 1s followed by a zero followed by a long string of 1s, e.g., 11111011111. Assume that the most recent preceding bit 1 for AMI has a negative voltage; and the most recent preceding bit 0 for pseudoternary has a negative voltage.

Draw the waveform for this sequence using

- a. NRZ-L
- b. Bipolar-AMI
- c. Pseudoternary

In NRZ-L, a negative voltage is used to represent binary one and a positive voltage is used to represent binary zero.

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Bipolar-AMI

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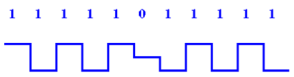
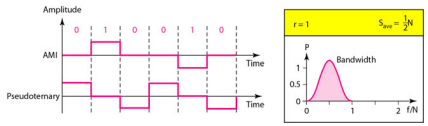
- One data element is represented by a zero voltage level, while the other alternates between negative and positive. AMI (Alternate Mark Inversion)
- – Binary 0 is represented by a neutral voltage, while Binary 1s are represented by alternating Positive and Negative





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Pseudoternary

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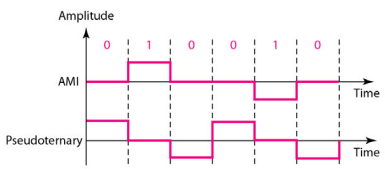
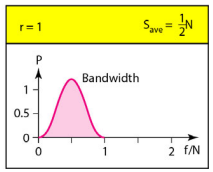
- A variation of AMI encoding is called **pseudoternary** in which the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.





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Question 4

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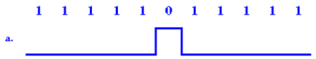
From the waveforms you draw in Q3, think about the following questions:

a. Does NRZ-L have the synchronization problem? If so, when?

有同步问题：很长时间内输出一个恒定的电压会导致信号漂移，失去同步



a. Yes, NRZ-L has the synchronization problem when there is a long sequence of 1s or 0s. Then, the output is a constant voltage over a long period of time. Under these situations, any drift between the clocks of transmitter and receiver will result in loss of synchronization.

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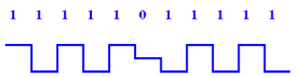
From the waveforms you draw in Q3, think about the following questions:
Does AMI have the synchronization problem? If so, when?



存在、若输入很长的一串0，
AMI可能信号漂移，
失去同步

Yes, AMI has the synchronization problem when a long sequence of 0s is present in the data

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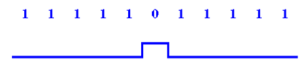
Cont'd

From the waveforms you draw in Q3, think about the following questions:
Does pseudoternary have the synchronization problem? If so, when?



可能、一串1，失去同步

Yes, pseudoternary has the synchronization problem when a long sequence of 1s is present in the data.

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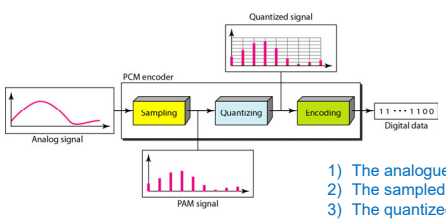
Question 5

- What are the processes of a PCM encoder?
- A PCM encoder has three processes, as shown in the figure below.

① Sampling
② Quantizing
③ Encoding



1) The analogue signal is sampled.
2) The sampled signal is quantized.
3) The quantized values are encoded as streams of bits

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Cont'd

- PCM and its variants are used in many other applications that require the conversion from analogue to digital:
  - Digital telephone
  - Digital Audio, digital speech
  - Digital Video
  - DVR, Blue-ray Disc and HD-DVD

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Sampling rate = 2 · fmax

更准确

一般来讲都会比2倍fmax还大，会oversampling一点。

Question 6

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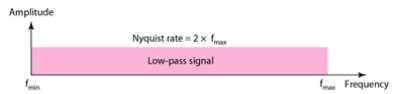
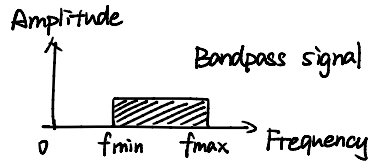
What is the Nyquist sampling rate for each of the following signals? Note that a low-pass signal indicates a signal that occupies the frequency range from 0Hz up to a certain cutoff, while a band-pass signal means a signal containing a band of frequencies not adjacent to zero frequency.

- A low-pass signal with bandwidth of 200 KHz?





According to the Nyquist theorem, to reproduce the original analogue signal, one necessary condition is that the sampling rate should be at least **twice the highest frequency** in the original signal.

a. In a low-pass signal, the minimum frequency is 0. Therefore, we have
fmax = 0 + 200 = 200 KHz. ⟹ fs = 2 * 200, 000 = 400,000 samples/s

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Question 6

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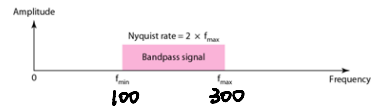
- A band-pass signal with bandwidth of 200 KHz if the lowest frequency is 100 KHz?



In a bandpass signal, the maximum frequency is equal to the minimum frequency plus the bandwidth. Therefore, we have
fmax = 100 + 200 = 300 KHz. ⟹ fs = 2 * 300,000 = 600,000 samples/s

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Question 7

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Consider an audio signal with spectral components in the range 300 to 3000 Hz. Assume that a sampling rate of 7000 samples per second will be used to generate a PCM signal.

- a. For SNR = 30 dB, what is the number of uniform quantization levels needed?
- b. What data rate is required?

a. $SNR_{dB} = 6.02\, n_b + 1.76\ dB$

$30 = 6.02\, n_b + 1.76$ dB

$n_b = (30-1.76)/6.02$

$n_b = 4.69$

Round it off and get that n = 5 bits. This yields 2^5= 32 quantization levels.

r=5 L = 2^5 = 32 个量化等级

即电平数

b. R = 7000 samples/s × 5 bits/sample = 35 Kbps

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N = S r = 7000×5 = 35kbps.

bit rate signal rate